

Learning Dynamics for High-Speed Tractor-Trailer Control

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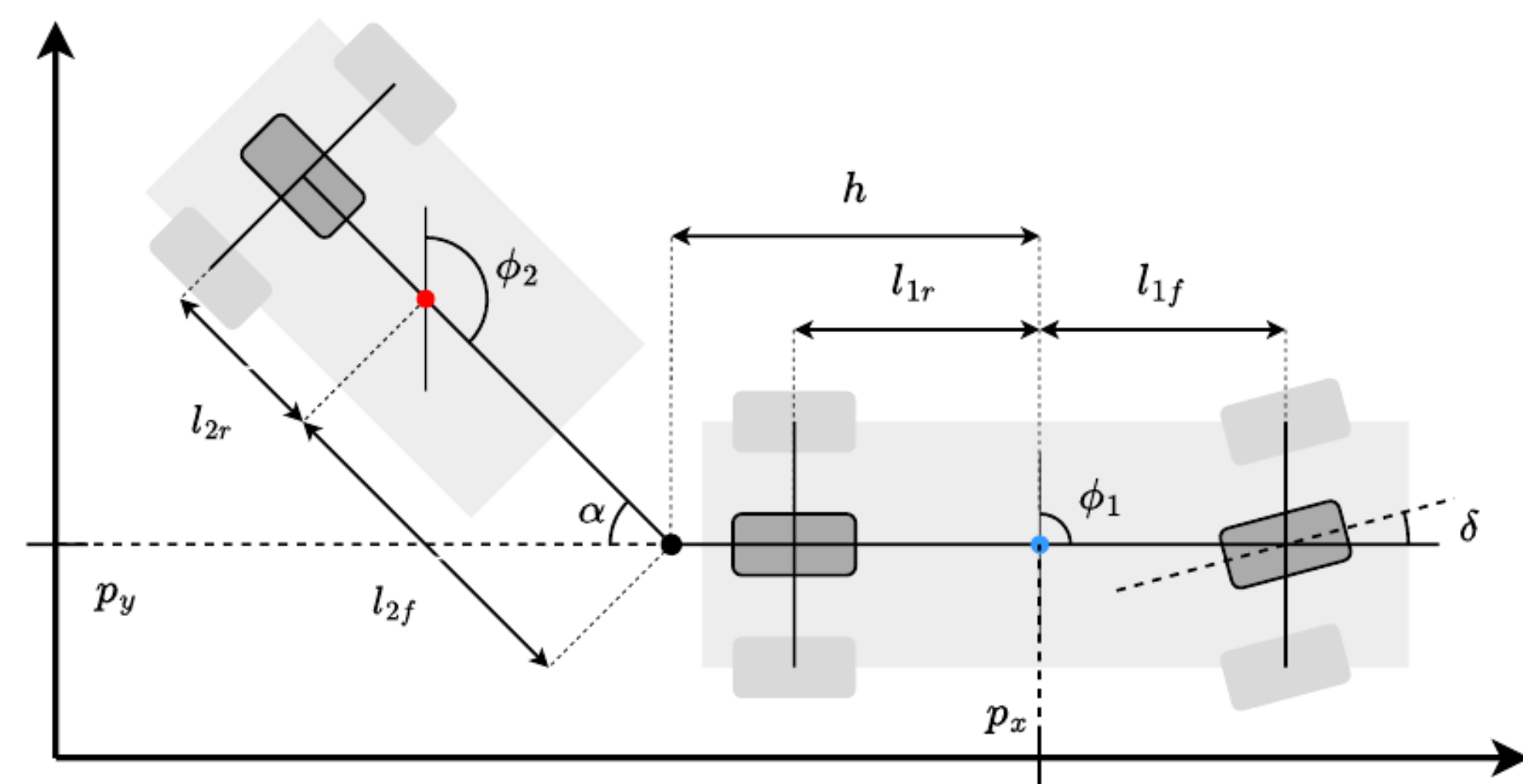
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Learned dynamics improve predictive control under changing road conditions.

High-speed tractor-trailer driving provides a demanding testbed.

Why this problem?

- Model mismatch: MPC decisions depend on the accuracy of predicted vehicle motion.
- Uncertainty: friction, terrain, and hidden vehicle states change the dynamics.
- Testbed: coupled tractor-trailer motion is nonlinear and unstable in reverse.



Rigid hitch coupling makes articulation error central to reverse control.

Learned model inside MPPI

1 Recent observations

Four-step history of state and controls

2 Neural dynamics

Predict accelerations and actuator states

3 MPPI rollouts

Score 500 sampled control sequences

4 Apply control

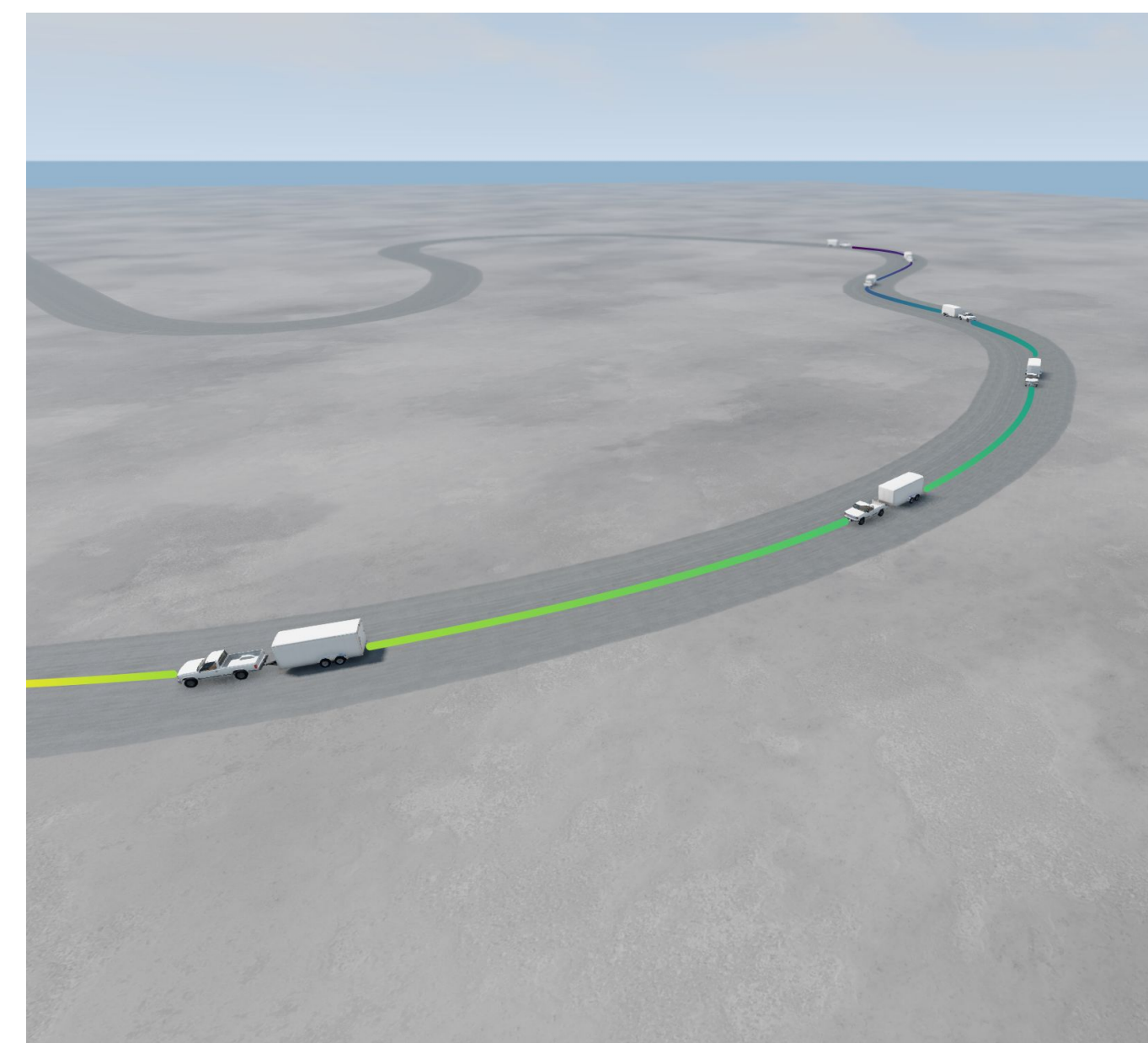
Observe the plant and repeat

Training: varied friction and speed. Slope and banking were withheld for evaluation.

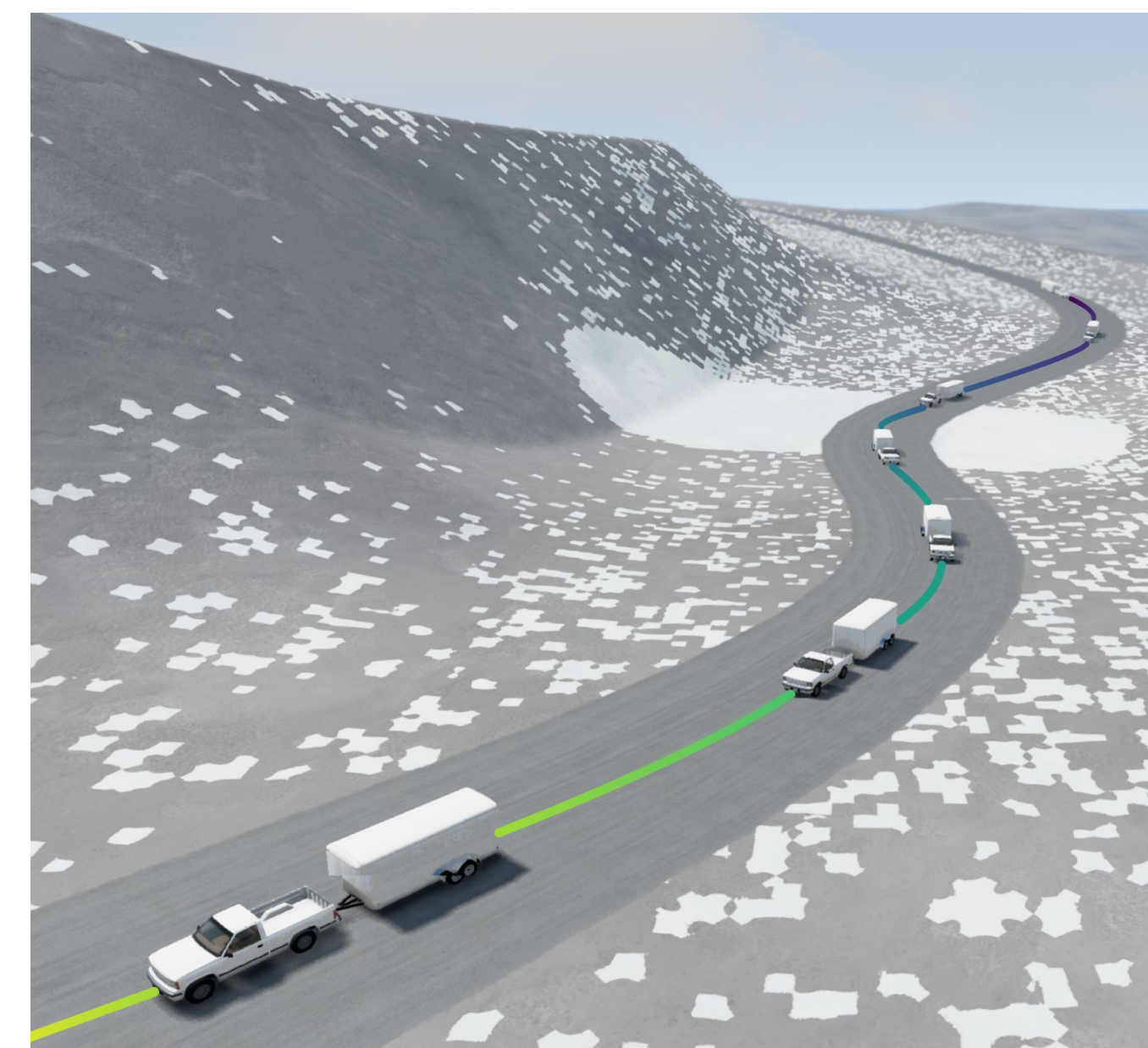
A demanding testbed

Four environments isolate changing friction and terrain; each configuration has nine trials.

ASPHALT · BASELINE

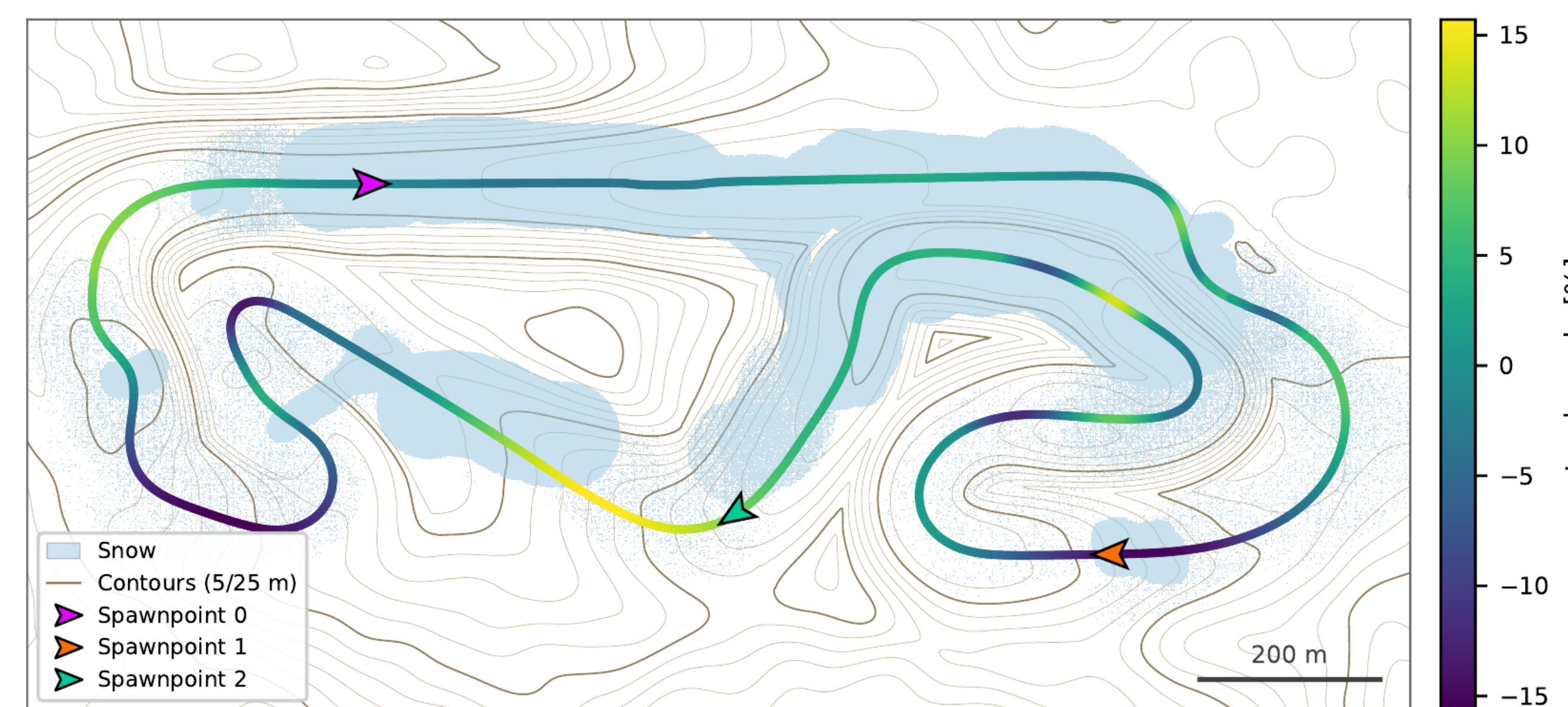


SNOW + SLOPE · COMBINED UNCERTAINTY



Representative reverse-driving trajectories with the learned controller.

What changes across the track?



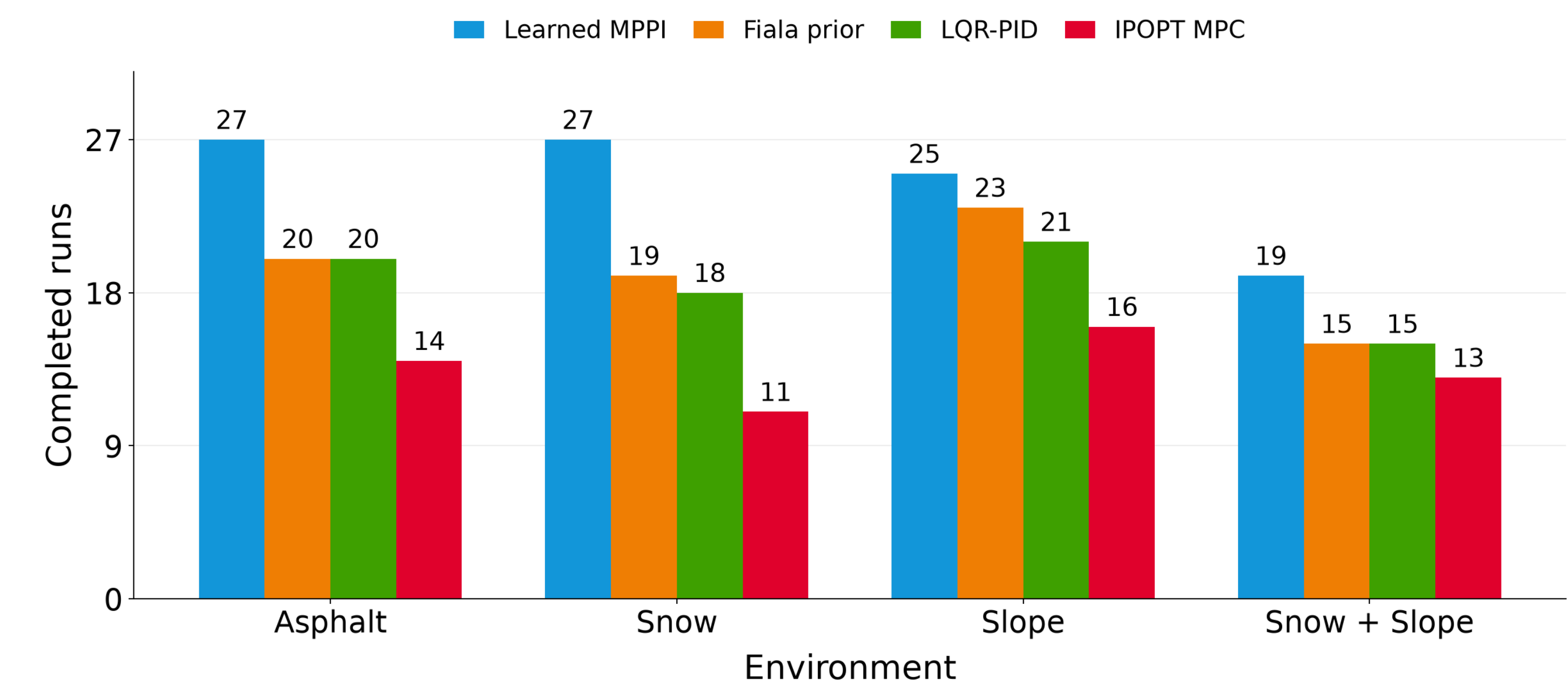
Friction varies in training. Sloped and banked terrain appear only in evaluation.

-30 / -50 km/h
REVERSE

+80 km/h
FORWARD

Closed-loop control

Completed 500-step runs across four environments and three speed targets.



98 / 108

completed with learned MPPI

62 / 72

reverse runs completed

Fiala-prior MPPI completed 77/108 overall and 41/72 in reverse.

Prediction accuracy

13 / 16

environment–direction–metric comparisons have the lowest RMSE with learned dynamics.

- Lowest reverse position RMSE in all four environments.
- Lowest forward hitch-angle RMSE in all four environments.
- Combined snow and slope still causes reverse-driving stalls.

Takeaway

Short-history learned dynamics improve MPPI control across tested conditions; the combined snow-and-slope case remains the hardest.